

一、(1) Stark 效应:

在强电场作用下原子发射的谱线分裂成几条的现象.

Zeeman 效应:

原子在外磁场中谱线发生分裂且偏振的现象.

$$(2) \vec{L} = \vec{r} \times \vec{p}$$

$$L_x = y p_z - z p_y$$

$$L_y = z p_x - x p_z$$

$$L_z = x p_y - y p_x$$

$$[L_x, L_y] = [y p_z - z p_y, z p_x - x p_z]$$

$$= y [p_z, z] p_x + x [z, p_z] p_y$$

$$= -i\hbar y p_x + i\hbar x p_y$$

$$= i\hbar (x p_y - y p_x) = i\hbar L_z$$

$$\text{同理 } [L_y, L_z] = i\hbar L_x$$

$$[L_z, L_x] = i\hbar L_y$$

$$(3) E_n^I \sim n^2 \quad \text{无限深势阱}$$

$$E_n^{II} \sim n \quad \text{谐振子}$$

$$E_n^{III} \sim \frac{1}{n^2} \quad \text{类氢原子}$$

(4) Schrödinger 方程:

$$\frac{\hat{p}^2}{2\mu} \psi(x) + V(x)\psi(x) = E \psi(x)$$

$$\hat{H} = \frac{\hat{p}^2}{2\mu} + V(\hat{x})$$

$$i\hbar \frac{d\hat{x}(t)}{dt} = [\hat{x}(t), \hat{H}(t)] = [\hat{x}(t), \frac{\hat{p}(t)^2}{2\mu} + V(\hat{x})]$$

$$= \frac{1}{2\mu} [\hat{x}(t), \hat{p}(t)^2]$$

$$= \frac{i\hbar}{\mu} \hat{p}(t)$$

$$i\hbar \frac{d\hat{p}(t)}{dt} = [\hat{p}(t), \hat{H}(t)] = [\hat{p}(t), \frac{\hat{p}(t)^2}{2\mu} + V(\hat{x}(t))]$$

$$= [\hat{p}(t), V(\hat{x}(t))]$$

$$= i\hbar (-\nabla V)$$

$$\Rightarrow \frac{d^2 \hat{x}(t)}{dt^2} = \frac{1}{\mu} \frac{d\hat{p}(t)}{dt} = -\nabla V(x)$$

$$\frac{d^2 \hat{p}(t)}{dt^2} = -\frac{d}{dt}(\nabla V) = -\frac{\partial(\nabla V)}{\partial x} \frac{d\hat{x}}{dt}$$

$$= -\frac{1}{\mu} \frac{\partial(\nabla V)}{\partial x} \hat{p}(t)$$

$$(5) \psi(a) = \psi(-a) = 0$$

$$(6) \vec{B} = (0, 0, B)$$

$$\hat{H} = \frac{1}{2\mu} \left[\hat{p} - \frac{q}{c} \vec{A}(\vec{r}, t) \right]^2 + q\Phi(\vec{r}, t)$$

$$\text{取 } \vec{A} = (-By, 0, 0)$$

$$\hat{H} = \frac{1}{2\mu} \left[(\hat{p}_x + \frac{qB}{c} y)^2 + \hat{p}_y^2 + \hat{p}_z^2 \right] + q\Phi(\vec{r}, t)$$

(7) 受激辐射是处于激发态的原子在外场作用下, 向低能态或激发态跃迁时辐射光子.

电偶极跃迁选择定则

$$\Delta l = \pm 1, \Delta m = 0, \pm 1$$

二、(1) 当 $|x| \geq b$ 时, Schrödinger 方程: (3)

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E \psi(x) = \frac{\hbar^2 k^2}{2m} \psi(x)$$

当 $|x| < b$ 时,

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) - V_0 \psi(x) = E \psi(x)$$

$$\text{令 } k' = \sqrt{k^2 + \frac{2mV_0}{\hbar^2}}$$

波函数:

$$\psi(x) = \begin{cases} e^{ikx} + R e^{-ikx}, & x \leq -b \\ A e^{ik'x} + B e^{-ik'x}, & -b < x < b \\ T e^{ikx}, & x \geq b \end{cases}$$

$$\text{由 } \begin{cases} \psi(b) = \psi(b^-) \\ \psi'(b^+) = \psi'(b^-) \end{cases} \quad \begin{cases} \psi(b^+) = \psi(-b^-) \\ \psi'(-b^+) = \psi'(-b^-) \end{cases} \text{ 得}$$

$$\begin{cases} A e^{ik'b} + B e^{-ik'b} = T e^{ikb} \\ ik' A e^{ik'b} - ik' B e^{-ik'b} = ik T e^{ikb} \end{cases}$$

$$\begin{cases} A e^{-ik'b} + B e^{ik'b} = e^{-ikb} + R e^{ikb} \\ ik' A e^{-ik'b} - ik' B e^{ik'b} = ik e^{-ikb} - ik R e^{ikb} \end{cases}$$

$$\Rightarrow R = \frac{(k^2 - k'^2)(1 - e^{4ik'b})e^{-2ikb}}{(k+k')^2 - (k-k')^2 e^{4ik'b}}$$

反射系数为

$$|R|^2 = \frac{2(1 - \cos 4k'b)}{\frac{1}{(k-k')^2} + \frac{1}{(k+k')^2} - 2 \cos 4k'b}$$

透射系数

$$|T|^2 = 1 - |R|^2 = \frac{\frac{1}{(k-k')^2} + \frac{1}{(k+k')^2} - 2}{\frac{1}{(k-k')^2} + \frac{1}{(k+k')^2} - 2 \cos 4k'b}$$

(2) 共振透射时, $|T| = 1$.

$$\cos 4k'b = 1$$

$$4k'b = 2n\pi, \quad n=1, 2, \dots$$

$$\Rightarrow k^2 = \frac{n^2 \pi^2}{4b^2} - \frac{2mV_0}{\hbar^2}, \quad n=1, 2, \dots$$

$$\text{三、} H' = \sum_{\alpha=1,2} \omega_{\alpha} |e\rangle \langle \alpha| + \text{H.c.}$$

$$\text{以 } |1\rangle, |2\rangle \text{ 为基, 令 } |e\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$H'_{11} = \langle 1 | H' | 1 \rangle$$

$$= \langle 1 | \sum_{\alpha=1,2} \omega_{\alpha} |e\rangle \langle \alpha| + \text{H.c.} | 1 \rangle$$

$$= \omega_1 \langle 1 | e \rangle \langle 1 | 1 \rangle + \langle 1 | \text{H.c.} | 1 \rangle$$

$$= c_1 \omega_1 + \cancel{(\text{H.c.})_{11}}$$

$$H'_{12} = \langle 1 | H' | 2 \rangle$$

$$= \langle 1 | \sum_{\alpha=1,2} \omega_{\alpha} |e\rangle \langle \alpha| + \text{H.c.} | 2 \rangle$$

$$= \omega_2 \langle 1 | e \rangle \langle 2 | 2 \rangle + \cancel{\langle 1 | \text{H.c.} | 2 \rangle}$$

$$= c_1 \omega_2 + \cancel{(\text{H.c.})_{12}}$$

$$H'_{21} = \langle 2 | H' | 1 \rangle$$

$$= c_2 \omega_1 + \cancel{(\text{H.c.})_{21}}$$

$$H'_{22} = c_2 \omega_2 + \cancel{(\text{H.c.})_{22}}$$

$$H' = \begin{pmatrix} c_1 \omega_1 & c_1 \omega_2 \\ c_2 \omega_1 & c_2 \omega_2 \end{pmatrix}$$

$$\text{由 } \begin{pmatrix} c_1 \omega_1 & c_1 \omega_2 \\ c_2 \omega_1 & c_2 \omega_2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = E_0^{(1)} \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\Rightarrow \begin{vmatrix} c_1 \omega_1 - E_0^{(1)} & c_1 \omega_2 \\ c_2 \omega_1 & c_2 \omega_2 - E_0^{(1)} \end{vmatrix} = 0$$

$$\Rightarrow (E_0^{(1)})^2 - (c_1 \omega_1 + c_2 \omega_2) E_0^{(1)} = 0$$

$$E_0^{(1)} = 0 \text{ 或 } E_0^{(1)} = c_1 \omega_1 + c_2 \omega_2$$

$$\text{由 } (H_0 + H') |e\rangle = E_1 |e\rangle$$

$$\Rightarrow \begin{pmatrix} E_0^{(0)} + c_1 \omega_1 & c_1 \omega_2 \\ c_2 \omega_1 & E_0^{(0)} + c_2 \omega_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = E_1 \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\Rightarrow c_1 \omega_1 + c_2 \omega_2 = E_1 - E_0^{(0)}$$

$$\text{当 } E_0^{(1)} = 0 \text{ 时, } E = E_0$$

$$\psi = \begin{pmatrix} \omega_2 \\ \omega_1 \end{pmatrix} = \omega_2 |1\rangle + \omega_1 |2\rangle$$

$$\text{当 } E_0^{(1)} = E_1 - E_0^{(0)} \text{ 时, } E = E_1$$

$$\psi = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = |e\rangle$$

四、类氢原子基态

$$\psi_{100}(z, \vec{r}) = \frac{1}{\sqrt{\pi}} \left(\frac{z}{a} \right)^{3/2} e^{-zr/a}$$

(1) 新态

$$\psi_{100}(z+1, \vec{r}) = \frac{1}{\sqrt{\pi}} \left(\frac{z+1}{a} \right)^{3/2} e^{-(z+1)r/a}$$

(2) 设 $t=0$ 时发生 β 衰变, $t>0$ 时

$$\psi(\vec{r}, t) = \sum_{nlm} C_{nlm} e^{-iE_{nlm}t/\hbar} \psi_{nlm}(z+1, \vec{r})$$

$$\psi(\vec{r}, 0) = \sum_{nlm} C_{nlm} \psi_{nlm}(z+1, \vec{r}) = \psi_{100}(z, \vec{r})$$

$$C_{nlm} = \int \psi_{nlm}^*(z+1, \vec{r}) \psi_{100}(z, \vec{r}) d\tau$$

β 衰变后, 仍处于基态概率为

$$|C_{100}|^2 = \left| \int \psi_{100}^*(z+1, \vec{r}) \psi_{100}(z, \vec{r}) d\tau \right|^2$$

$$= \left| \int \frac{[z(z+1)]^{3/2}}{\pi a^3} e^{-(2z+1)r/a} d\tau \right|^2$$

$$= \frac{[z(z+1)]^3}{\pi^2 a^6} \left| 4\pi \int_0^\infty e^{-(2z+1)r/a} r^2 dr \right|^2$$

$$= \frac{32[z(z+1)]^3}{(2z+1)^6}$$

$$\text{五、(1)} \quad \hat{L}_z Y_{lm} = m\hbar Y_{lm}$$

$$t>0 \text{ 时, } \psi(t) = a e^{-iE_{11}t/\hbar} Y_{11} + b e^{-iE_{10}t/\hbar} Y_{10}$$

$$= a e^{-i\mu_B t} Y_{11} + b Y_{10}$$

$\hat{L}_z$ 可能值 $\hbar, 0$

$$\text{概率分别为 } |\langle Y_{11} | \psi(t) \rangle|^2 = |a|^2$$

$$|\langle Y_{10} | \psi(t) \rangle|^2 = |b|^2$$

$$(2) \text{ 取 } Y_{11}, Y_{10}, Y_{1,-1} \text{ 为基 } \psi(0) = \begin{pmatrix} a \\ b \\ 0 \end{pmatrix}$$

$$\psi(t) = \begin{pmatrix} a e^{-i\mu_B t} \\ b \\ 0 \end{pmatrix}$$

$$L_x = \frac{1}{2}(L_+ + L_-)$$

$$L_x Y_{11} = \frac{1}{2} \sqrt{1 \times 2 - 1 \times 0} \hbar Y_{10} = \frac{\sqrt{2}}{2} \hbar Y_{10}$$

$$L_x Y_{10} = \frac{1}{2} \sqrt{1 \times 2 - 0 \times 1} \hbar Y_{11} = \frac{\sqrt{2}}{2} \hbar Y_{11} \dots$$

$$\hat{L}_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$L_x = \hbar, \varphi_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}, \quad L_x = 0, \varphi_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$L_x = -\hbar, \varphi_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}$$

$$\psi(t) = \begin{pmatrix} a e^{-i\mu_B t} \\ b \\ 0 \end{pmatrix} = \left(\frac{a}{2} e^{-i\mu_B t} + \frac{\sqrt{2}}{2} b \right) \varphi_+ + \left(\frac{a}{2} e^{-i\mu_B t} - \frac{\sqrt{2}}{2} b \right) \varphi_- + \frac{a}{\sqrt{2}} e^{-i\mu_B t} \varphi_0$$

$$L_x \text{ 取 } \hbar \text{ 概率为 } |\langle \varphi_+ | \psi(t) \rangle|^2 = \frac{|a|^2}{4} + \frac{|b|^2}{2} + \frac{\sqrt{2}}{2} |a||b| \cos \mu_B t$$

$$L_x \text{ 取 } 0 \text{ 概率为 } |\langle \varphi_0 | \psi(t) \rangle|^2 = \frac{|a|^2}{2}$$

$$L_x \text{ 取 } -\hbar \text{ 概率为 } |\langle \varphi_- | \psi(t) \rangle|^2 = \frac{|a|^2}{4} + \frac{|b|^2}{2} - \frac{\sqrt{2}}{2} |a||b| \cos \mu_B t$$

六、一维无限深势阱:

$$\psi_n(x) = \sqrt{\frac{1}{b}} \sin \frac{n\pi x}{2b}$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{8mb^2}, \quad n=1, 2, 3, \dots$$

(1) 基态:

$$\psi = \psi_1(x_1) \psi_1(x_2) |00\rangle$$

$$E_0 = E_{11} = \frac{\pi^2 \hbar^2}{4mb^2}$$

第一激发态:

$$\psi = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2)] |00\rangle$$

$$\psi = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) - \psi_2(x_1) \psi_1(x_2)] \left. \begin{array}{l} |11\rangle \\ |10\rangle \\ |1-1\rangle \end{array} \right\}$$

$$E_1 = E_{12} = \frac{5\pi^2 \hbar^2}{8mb^2}$$

(2) 加入均匀磁场B时,

$$H' = -\vec{\mu} \cdot \vec{B} = \frac{eB}{2m} S$$

$$S|00\rangle = 0$$

$$S|11\rangle = \hbar|11\rangle$$

$$S|10\rangle = 0$$

$$S|1-1\rangle = -\hbar|1-1\rangle$$

基态, $E_0 = \frac{\pi^2 \hbar^2}{4mb^2}$

第一激发态:

$$E_1 = \left\{ \begin{array}{l} \frac{5\pi^2 \hbar^2}{8mb^2} \\ \frac{5\pi^2 \hbar^2}{8mb^2} + \frac{eB\hbar}{2m} \\ \frac{5\pi^2 \hbar^2}{8mb^2} - \frac{eB\hbar}{2m} \end{array} \right.$$

临界 $B_c: \frac{5\pi^2 \hbar^2}{8mb^2} - \frac{eB\hbar}{2m} = \frac{\pi^2 \hbar^2}{4mb^2}$

$$B_c = \frac{3\pi^2 \hbar}{4b^2 e}$$

基态变为 $\psi = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) - \psi_2(x_1) \psi_1(x_2)] |1-1\rangle$